

Factor Models of Interest Rate Curves

Homework Assignment #1

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Abstract

We study cross-sectional and dynamic factor models of the U.S. Treasury yield curve using the Nelson–Siegel (NS) framework. Starting from a closed-form derivation of zero-coupon yields, we build a daily panel from the Gürkaynak–Sack–Wright (2007) dataset and calibrate the NS model via three approaches to choosing the decay parameter λ : grid search with conditional OLS, a fixed λ selected by in-sample minimization of median RMSE, and full nonlinear joint optimization. We document that the grid search and nonlinear approaches deliver near-identical fit quality (median RMSE $\approx 0.0016\%$), while the fixed- λ approach incurs material deviations during monetary-policy transitions such as the 2018 flattening cycle and the 2022–2023 inversion. As an extension, we impose a VAR(1) time-series structure on the NS factors and estimate them via the Kalman filter, showing that filtered factor paths are substantially smoother and eliminate the numerical outliers that contaminate the daily OLS estimates. We then evaluate the model in forecasting and trading applications, finding that DNS–Kalman forecasts do not outperform random-walk or AR(1) benchmarks for yield changes at daily, weekly, or monthly horizons (Part E).

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1. Closed-Form Yields (Part A)

The Nelson–Siegel yield curve expresses the zero-coupon yield at maturity T as a linear combination of three factor loadings (see Appendix A for the full derivation):

$$Y(T) = \beta_0 + \beta_1 A_1(T; \lambda) + \beta_2 A_2(T; \lambda), \quad (1)$$

where the loadings are

$$A_1(T; \lambda) = \frac{1 - e^{-\lambda T}}{\lambda T}, \quad A_2(T; \lambda) = \frac{1 - e^{-\lambda T}}{\lambda T} - e^{-\lambda T}. \quad (2)$$

1.1. Economic Interpretation

To interpret the NS factors, figure 1 plots the loadings A_1 and A_2 as functions of maturity for a representative $\lambda = 1$. A_1 is a monotone, decaying function that starts at 1 and asymptotes to zero; A_2 is a hump-shaped function that starts at zero, peaks at medium maturities, and decays back to zero at long maturities.

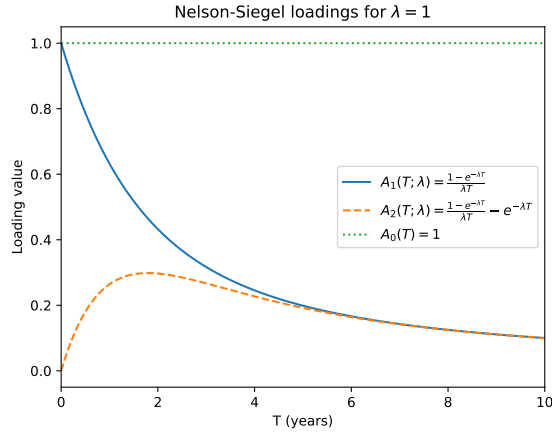


Figure 1: Nelson–Siegel loadings A_1 and A_2 for $\lambda = 1$.

- **Level** (β_0): since as $T \rightarrow \infty$, $A_1(T; \lambda) \rightarrow 0$ and $A_2(T; \lambda) \rightarrow 0$,

$$\lim_{T \rightarrow \infty} Y(T) = \beta_0.$$

This is the long-run level of the curve.

- **Short end / slope** (β_1): as $T \rightarrow 0$, $A_1(T; \lambda) \rightarrow 1$ and $A_2(T; \lambda) \rightarrow 0$, so

$$\lim_{T \rightarrow 0} Y(T) = \beta_0 + \beta_1.$$

If $\beta_1 > 0$, the curve is usually downward-sloping: the short end ($\beta_0 + \beta_1$) starts above the long-run level (β_0) and decays down to it. If $\beta_1 < 0$, the curve is usually upward-sloping: the short end starts below the long-run level and rises up to it.

- **Curvature** (β_2): for both $T \rightarrow 0$ and $T \rightarrow \infty$, $A_2(T; \lambda) \rightarrow 0$. However, A_2 peaks at medium maturities, so β_2 controls mid-maturity curvature (a hump if $\beta_2 > 0$; a dip if $\beta_2 < 0$).
- **Scale** (λ): increasing λ makes the exponentials decay faster, concentrating slope/curvature effects at shorter maturities. A rough curvature timescale is $T \approx 1/\lambda$.

2. Data Construction (Part B)

We construct a daily panel of zero-coupon Treasury yields from January 1, 2016 to early 2026 using the Gürkaynak, Sack, and Wright (2007) dataset published by the Federal Reserve Board¹. The dataset provides daily Svensson (1994) parameters $(\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2)$ fitted to the outstanding Treasuries daily. This model has more parameters and greater flexibility than the NS model, so it can be used to construct a clean panel of zero-coupon yields at arbitrary maturities. As the Svensson fit matches the observed Treasury market data closely (error is below 1 basis point, as noted in the paper), the implied zero-coupon yield curve is a reliable synthetic discount curve constructed from those bond prices.

Yields at the eight target maturities $T \in \{0.25, 0.5, 1, 2, 3, 5, 7, 10\}$ years are obtained by evaluating the Svensson formula with the daily parameter estimates. Sub-year maturities are purely model-implied rather than directly observed, but this ensures consistent continuous-compounding conventions across the full panel. After dropping days with missing parameters, the panel contains 2,524 trading days. Using zero-coupon (rather than constant-maturity par) yields avoids the coupon bias that arises when fitting a discount-curve model: par yields conflate discounting across horizons and would introduce systematic errors in the NS calibration.

The evaluated yields are plotted in Figure 2. The panel captures the full range of yield curve regimes during the sample, including the

- 2016–2019 hiking cycle with a progressive flattening of the curve,
- 2020 COVID shock with a flat curve and near-zero rates,
- 2022–2023 hiking cycle with inversion and subsequent steepening,
- 2024–2025 steepening cycle as the Fed pivots to easing, short-end yields decline and the long end remains elevated.

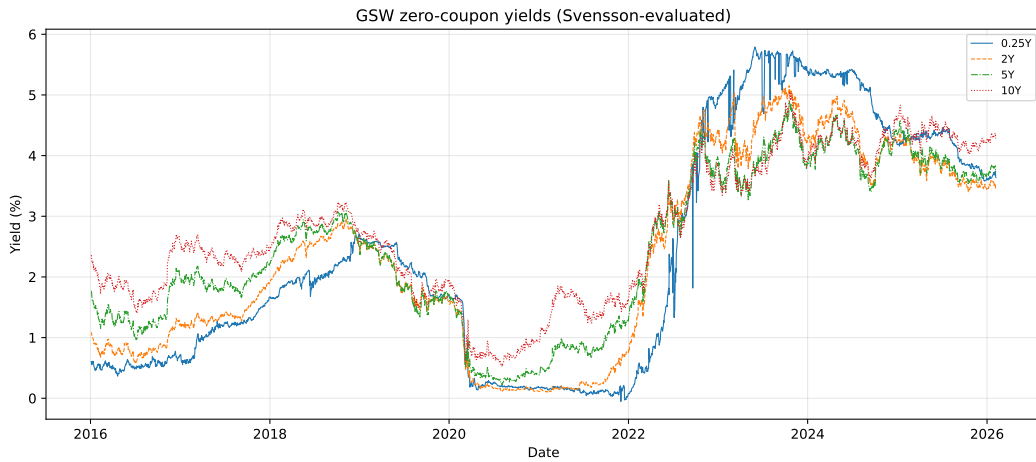


Figure 2: Daily zero-coupon yields at four selected maturities from the GSW dataset. Note that maturity 0.25 years is purely model-implied (extrapolated from the Svensson fit). While the rest of the maturities are directly observed, they are still model-implied in the sense that they are derived from the Svensson parameters rather than directly from par yields.

¹<https://www.federalreserve.gov/econres/feds/the-us-treasury-yield-curve-1961-to-the-present.htm>

3. Cross-Sectional Calibration (Part C)

For each day t we minimise the equally-weighted sum of squared residuals:

$$\min_{\beta_0, \beta_1, \beta_2, \lambda} \sum_{i=1}^8 (Y_t(T_i) - \hat{Y}(T_i; \beta_0, \beta_1, \beta_2, \lambda))^2. \quad (3)$$

For fixed λ the problem is linear in $(\beta_0, \beta_1, \beta_2)$ and solved in closed form by OLS.

3.1. Three Approaches to Choosing λ

(a) Grid search with conditional OLS. For each of 120 log-spaced values $\lambda \in [0.05, 5.0]$ we solve the OLS problem and record the SSE; the day's λ is the minimiser. This procedure is run independently for every trading day, so λ varies over time.

(b) Fixed λ by in-sample grid search. We select a single global λ^* by minimising the *median* in-sample RMSE across all 2,524 days:

$$\lambda^* = \arg \min_{\lambda} \text{median}_t \left\{ \sqrt{\frac{1}{8} \sum_i (Y_t(T_i) - \hat{Y}(T_i; \hat{\beta}_t(\lambda), \lambda))^2} \right\}. \quad (4)$$

The median is used instead of the mean to reduce the influence of outlier days.² This selects $\lambda^* = 0.41$, which places the NS curvature hump at roughly $\tau^* \approx 1/\lambda^* \approx 2.4$ years. Since the minimizer is unchanged by positive monotone rescalings of the objective, one could equivalently minimize median SSE (i.e. without the square root and the division by 8), but we report RMSE because it is directly interpretable in yield units.

Figure 3 plots the RMSE as a function of λ ; the minimum is unique and well-defined, the curve is smooth and convex around the minimum, so the optimal λ is well-identified and not sensitive to small perturbations.

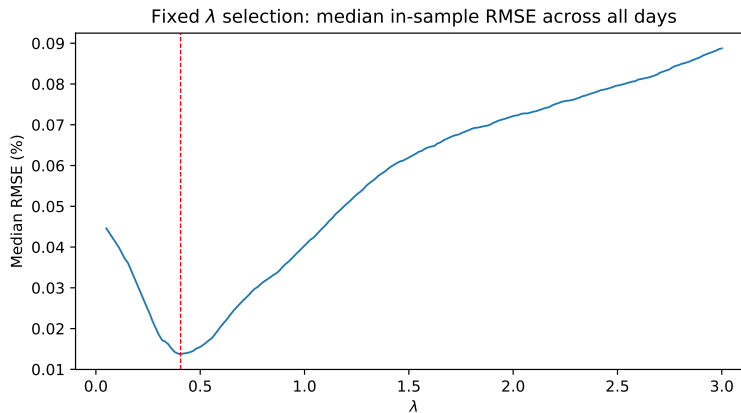


Figure 3: Fixed λ median RMSE across all days. The minimum is broad and flat, so the fit quality is not very sensitive to the exact choice of $\lambda \approx 0.41$.

²We minimize the median daily RMSE, rather than the pooled sum of squared residuals, because the goal is to choose a single fixed decay parameter that performs well on a typical day. A pooled SSE criterion would overweight a small number of stressed or numerically unstable days, whereas the median is robust to such outliers.

(c) Full nonlinear optimisation. We jointly optimize over $(\beta_0, \beta_1, \beta_2, \lambda)$ using L-BFGS-B with $\lambda \in [0.01, 10]$. L-BFGS-B is efficient for smooth objectives with box constraints and the NS objective is differentiable in all four parameters. Since L-BFGS-B is a local method, we warm-start from the grid-search optimum from (a) to avoid finding local minima.

3.2. Fit Quality

Table 1 summarises in-sample RMSE across all days. Methods (a) and (c) are near-identical, implying that the grid-search resolution is sufficiently fine and leaves little room for the nonlinear optimiser to improve. Method (b) incurs a much higher median and maximum RMSE, mostly driven by stressed-regime days.

Table 1: In-sample RMSE (%) summary across 2,524 trading days.

Method	Mean	Median	Max	Mean λ	Std λ
(a) Grid search	0.0053	0.0016	0.0615	0.64	0.57
(b) Fixed $\lambda = 0.41$	0.0261	0.0137	0.5009	0.41	0.00
(c) Nonlinear opt	0.0053	0.0015	0.0615	0.64	0.56

Figure 4 shows the RMSE time series. The grid search and nonlinear optimisation paths overlap almost exactly. The fixed- λ method tracks them during calm regimes but diverges sharply during stress episodes, especially during the 2022–2023 hiking cycle (RMSE approaching 0.2%, roughly 10 times above the adaptive methods). At the second half of 2018 (inset), all methods show elevated RMSE coinciding with the Fed’s September and December rate hikes and the associated curve flattening. The RMSE distribution in Figure 5 confirms that fixed λ produces a much heavier tail of high-RMSE days.

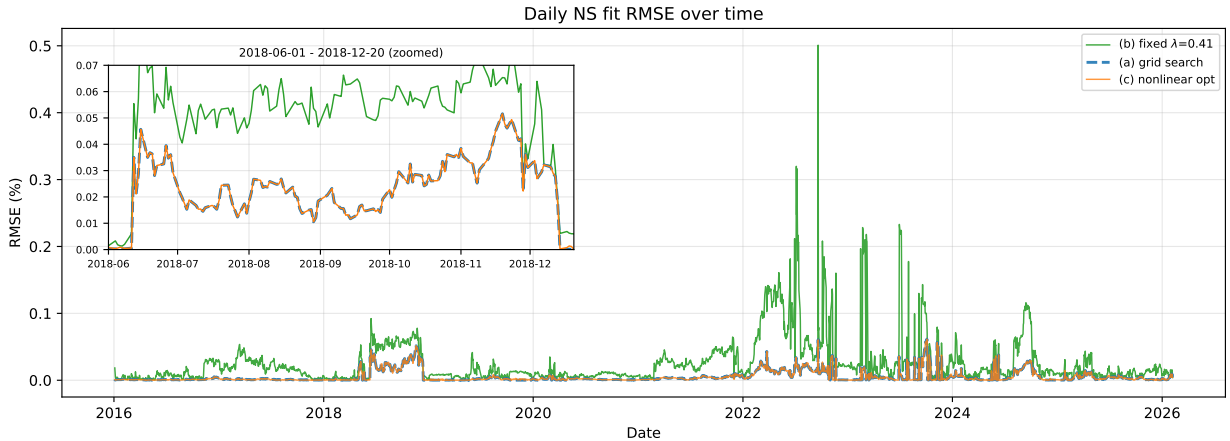


Figure 4: Time series of in-sample RMSE for the three λ approaches. The nonlinear optimisation and grid search methods have overlapping RMSE paths. The inset zooms in on the 2018-06-01 to 2018-12-20 period, showing a spike in RMSE for all methods.

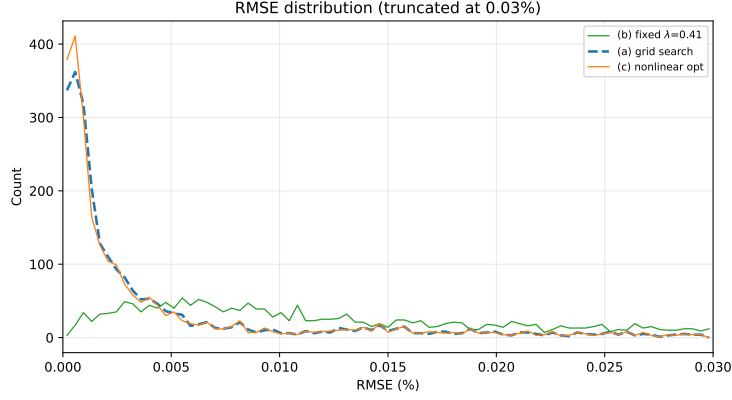


Figure 5: Distribution of in-sample RMSE across days for the three λ approaches. The fixed λ method has a much heavier tail of high-RMSE days than the adaptive methods.

3.3. Curve Fits on Selected Dates

Figure 6 shows observed-vs-fitted curves on four dates spanning calm and stressed regimes.

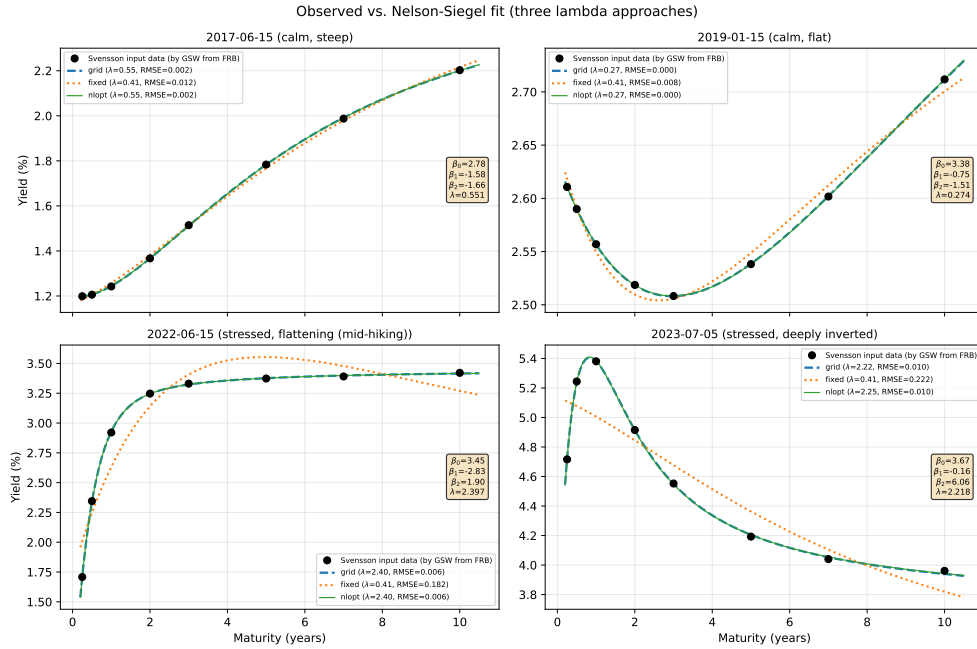


Figure 6: Svensson fit (input data) vs. Nelson–Siegel fit (three λ approaches) on four selected dates.

In calm periods (2017-06-15, 2019-01-15) all three methods fit well; the fixed- λ penalty is small ($\text{RMSE} \leq 0.012\%$). On 2022-06-15, during the mid-hiking-cycle flattening, the adaptive methods select $\lambda = 2.40$ and achieve $\text{RMSE} = 0.006\%$, while the fixed $\lambda = 0.41$ yields $\text{RMSE} = 0.182\%$ — thirty times larger. On 2023-07-05, during the peak inversion, the adaptive methods select $\lambda \approx 2.22$ – 2.25 ($\text{RMSE} = 0.010\%$), while the fixed λ fails to replicate the inverted shape ($\text{RMSE} = 0.222\%$).

The underlying cause is visible in Figure 7: the optimal λ shifts far from 0.41 during stressed

regimes, and the fixed loading structure simply cannot accommodate the changed curve shape. In the first half of 2018, the nonlinear optimisation shows instability in λ while the grid search remains more stable; in the second half, both methods produce elevated but stable values, coinciding with the September and December rate hikes.

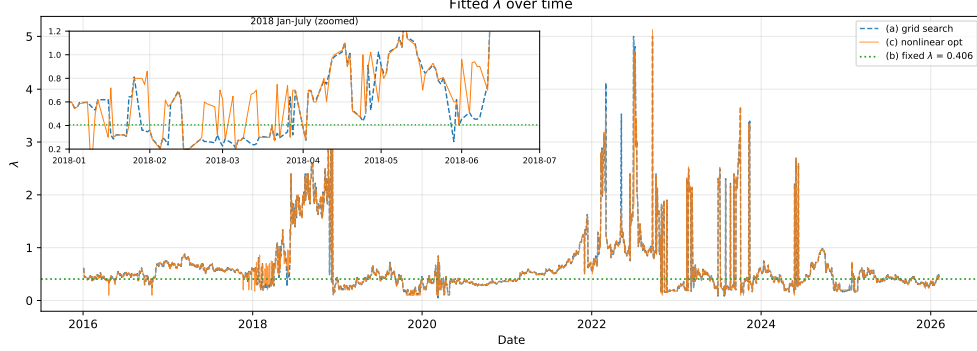


Figure 7: Time series of λ for the three fitting methods. Grid search and nonlinear optimisation largely overlap; the fixed method is a horizontal line at $\lambda = 0.41$.

3.4. NS Factors and Their Stability

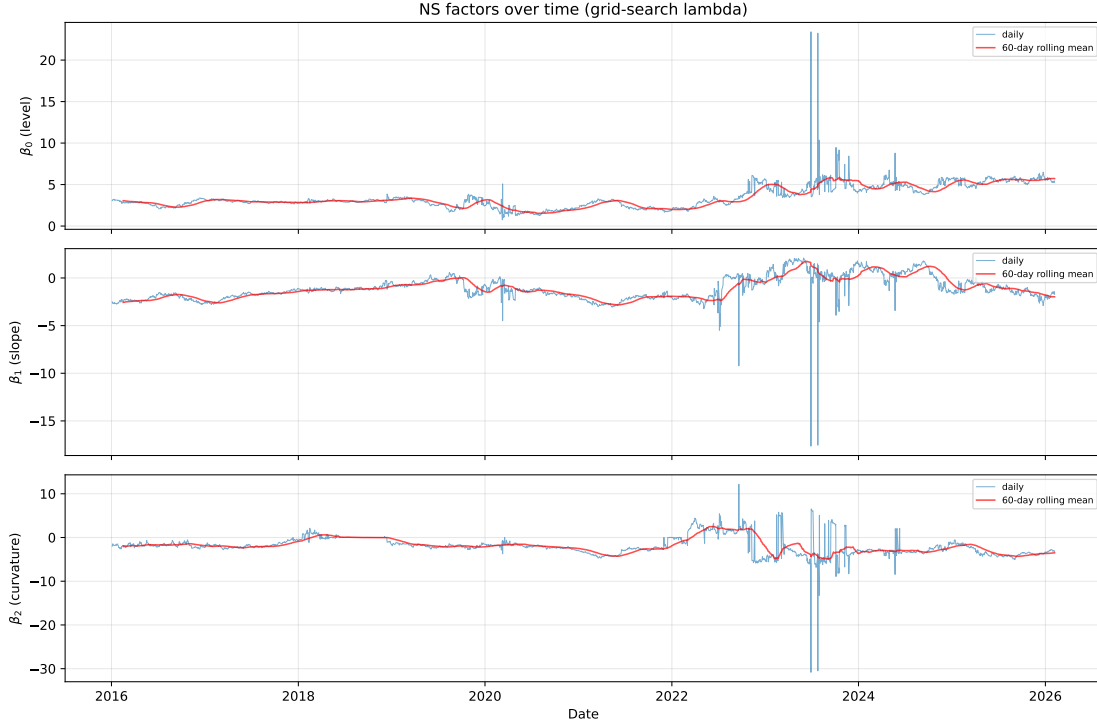


Figure 8: NS factors $(\beta_0, \beta_1, \beta_2)$ over time estimated via grid-search λ .

Figure 8 plots the daily NS factors over the full sample, with a 60-day lookback rolling mean overlaid. Table 2 reports the corresponding summary statistics for the grid-search factors.

Table 2: NS factor summary statistics (grid-search λ , 2016–2026).

	Mean	Median	Std	Min	Max
β_0 (level)	3.44	3.04	1.39	0.74	23.41
β_1 (slope)	-1.11	-1.32	1.27	-17.64	2.09
β_2 (curvature)	-1.93	-2.09	2.15	-30.84	12.18

- β_0 (level) averages 3.44% and remains relatively stable among the three factors. The rolling mean rises smoothly from around 2% in 2016 to a peak near 5–6% in 2023–2024, reflecting the Fed hiking cycle.

Some sharp upward spikes are visible around 2022–2023, where daily estimates briefly reach values above 20%. These are clear numerical outliers caused by the NS objective function admitting two near-equivalent local minima on those days, with one solution being economically implausible but having better fit than the sensible solution. We investigate this instability and propose two solution within this in subsection B, but the Kalman filter in Part D provides a more principled solution by regularising the factor paths through the VAR(1) time-series structure.

- β_1 (slope) is negative on average, consistent with an upward-sloping curve for most of the sample. It briefly turned positive during the 2022–2023 inversion, as the rolling mean crossing zero.

The daily series shows a pronounced downward spike near 2023, reaching approximately -15% . As they occur during the same period as the β_0 outliers, these slope spikes are likely part of the same fitting instability rather than a genuine market signal.

- β_2 (curvature) averages -1.93% and is the most volatile factor, with daily values ranging from below -30% to above $+10\%$. The rolling mean mostly remains close to zero but dips noticeably negative during 2022–2024, reflecting the pronounced stressed curvature of the inverted curve.

The extreme daily spikes across all three factors are concentrated in the 2022–2023 period.

Figure 9 shows rolling 60-day volatility during 2016–2021 has median below 0.35% for all three factors, though occasional spikes reach 0.7–1.0%. During 2022–2023 the median volatility rises by a factor of 4–7 $\times$ (e.g. β_2 jumps from median 0.31 to 2.24), then reverts after 2024, reinforcing that the instability was regime-driven. β_1 is largely uncorrelated with β_0 and β_2 (0.12 and 0.02 respectively), implying that level and slope movements are distinct. β_0 and β_2 exhibit moderate negative correlation (-0.51). Level and curvature factors moving in opposite directions suggests that when the long-run level rises, estimated curvature tends to fall. This is a known feature of the NS parameterisation — a higher β_0 mechanically shifts the entire curve up, reducing the apparent hump relative to the long end, which β_2 compensates for by becoming more negative.

4. Dynamic Nelson–Siegel via Kalman Filter (Part D1)

The two-step OLS procedure of Parts B and C estimates NS factors independently at each date, ignoring a time-series structure. Part D imposes a VAR(1) on the latent factors by casting the DNS model as a linear state-space system estimated with the Kalman filter, simultaneously regularizing

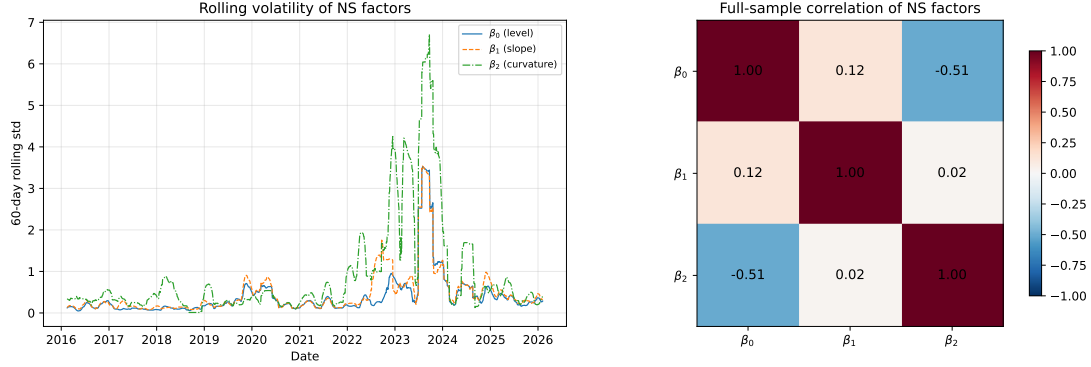


Figure 9: Diagnostic plots for the NS factors estimated via grid-search λ . *Left*: 60-day rolling volatility (standard deviation) of β_0 , β_1 , and β_2 . *Right*: Full-sample pairwise correlation matrix of the three factors.

factor estimates and producing out-of-sample forecasting. The measurement and state equations are given in Appendix C; the Kalman filter recursions and VAR diagnostics are in Appendices D and E.

4.1. Factor Estimates: DNS vs. OLS vs. Nonlinear

Figure 10 compares the three factor series over the full sample. The major distinction can be seen in the behaviour around summer 2023, as an example, where both OLS and nonlinear methods produce β_0 values above 20 (maximum 23.40 and 16.55 respectively), with corresponding extreme spikes in β_1 and β_2 . These are result of the NS objective function admitting two near-equivalent local minima on those days. The Kalman filter resolves this because the VAR prior implies β_0 should lie near 4–5, making the outlier solution incompatible with the predicted state, so the filtered result remains stable and coherent.

4.2. In-Sample Fit

Table 3: In-sample RMSE summary. All three methods are evaluated using the same fixed $\lambda^* = 0.406$ loadings matrix. The OLS and nonlinear factors were originally estimated with day-specific λ values, so their RMSE here reflects the mismatch between the λ at which they were fitted and the λ at which they are evaluated.

Method	Median (%)	Mean (%)	Max (%)
OLS grid search	0.1296	0.1903	4.3532
Nonlinear opt	0.1338	0.1950	4.5065
Kalman filtered	0.0137	0.0261	0.5009

Table 3 reports in-sample RMSE recomputed under the same fixed $\lambda^* = 0.406$ for all methods. The Kalman filter achieves $10\times$ lower median RMSE than either two-step method (0.0137% vs. 0.1296%). Especially on outlier days, the two-step methods converged to poor very local solutions with high fitting error; the filter, anchored to the sensible solution by the VAR prior, achieves better fit and stability simultaneously. The maximum daily RMSE falls from 4.35% (OLS) to 0.50% (Kalman), an $8.7\times$ reduction.

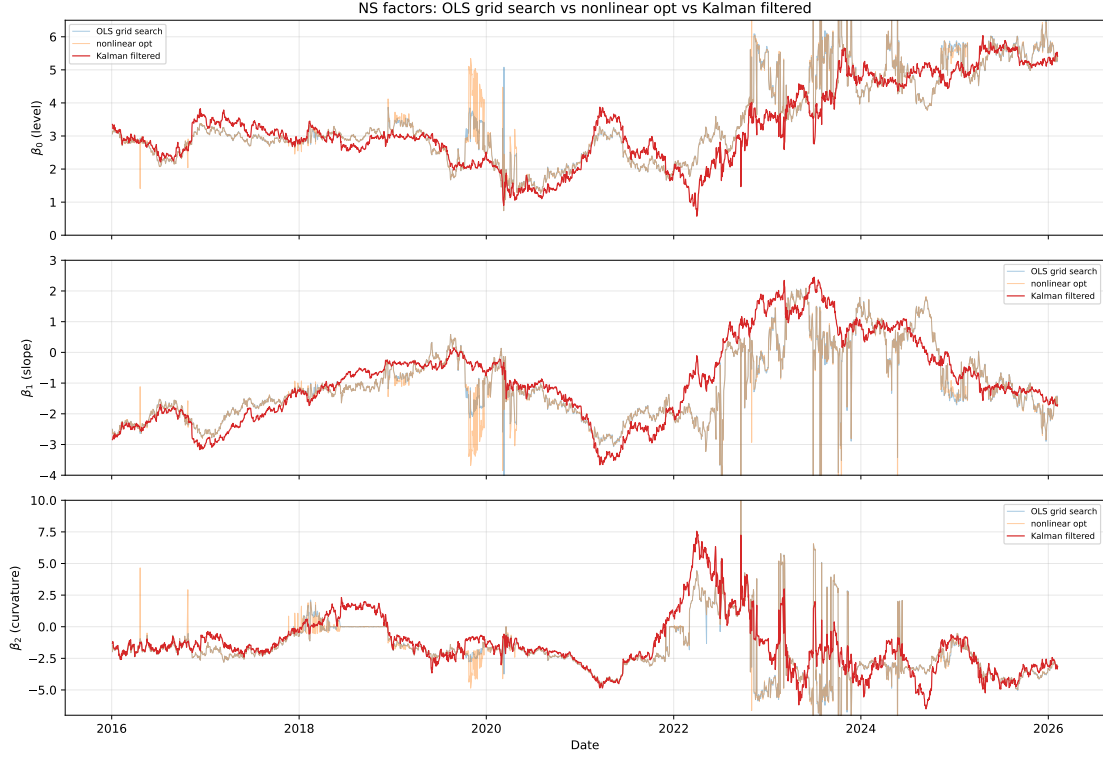


Figure 10: NS factors estimated by three methods. OLS and nonlinear series (faint) spike to implausible values in summer 2023; the Kalman filtered series (red) remains economically coherent throughout.

4.3. Out-of-Sample Forecasting

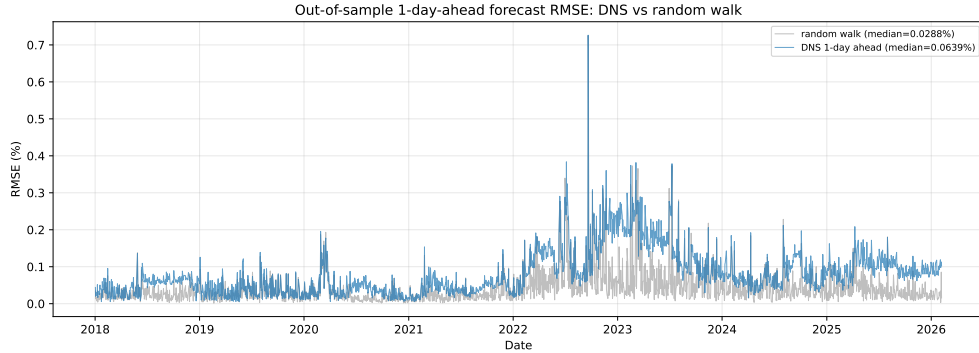


Figure 12: Daily out-of-sample 1-day-ahead forecast RMSE for the DNS Kalman filter and random walk benchmark (January 2018 – February 2026). The DNS series lies consistently above the random walk, with the gap widening markedly during the 2022–2023 rate-hiking cycle.

We evaluate one-day-ahead forecasting over 2,024 out-of-sample days (January 2018 – February 2026) using a strictly causal rolling design: at each day t , the Kalman predict step $\hat{\mathbf{X}}_{t|t-1} = \mathbf{c} + \Phi \hat{\mathbf{X}}_{t-1|t-1}$ is applied before observing $\mathbf{Y}_t$, and VAR parameters are re-estimated every 60 days on expanding

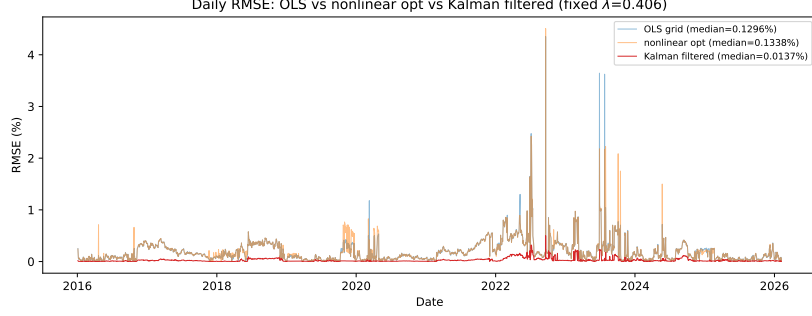


Figure 11: Daily in-sample RMSE for OLS grid search, nonlinear optimisation, and Kalman filtered factors (all evaluated at fixed $\lambda^* = 0.406$). The Kalman series (red) remains below 0.5% throughout; OLS and nonlinear spike to over 4% on outlier days concentrated in 2022–2024.

data. The benchmark is the random walk (RW): $\hat{\mathbf{Y}}_t^{\text{RW}} = \mathbf{Y}_{t-1}$.

Table 4: Out-of-sample 1-day-ahead forecast RMSE

Method	Median RMSE (%)	Cumulative SSE ratio
Random walk	0.0288	1.000
DNS Kalman filter	0.0639	2.5281

The DNS Kalman filter is outperformed by the random walk at this horizon (cumulative SSE ratio 2.53, Table 4). The dominant near-unit-root eigenvalue (Appendix E) could explain why: at a one-day horizon the VAR predicts only a negligible pull toward the long-run mean, barely distinguishable from a random walk but with added estimation error. The gap widens further during the 2022–2023 rate-hiking cycle, when yields trended strongly away from their historical mean and the VAR’s mean-reversion prior was adverse.

These results should be interpreted alongside the in-sample findings. The filter provides outlier-free, economically coherent factors with substantially lower fitting error. This is valuable for curve interpolation, risk factor extraction, and derivatives pricing. However, those factors cannot forecast yield level better than the random walk at the 1-day horizon.

5. Applications: Forecasting and Trading (Part E)

We use the DNS–Kalman model to forecast *changes* in 2-, 5-, and 10-year zero-coupon yields over horizons of 1, 5, and 21 business days. Forecasts are generated in a strictly out-of-sample manner using a rolling three-year estimation window, with the VAR(1) parameters and Kalman filter re-estimated at each forecast origin. We compare three models: a random-walk benchmark for yield changes (imposing $\Delta Y_t = 0$), a univariate AR(1) for each maturity, and the DNS–Kalman model, which produces multi-step factor forecasts that are mapped into yields via the Nelson–Siegel loadings at fixed λ^* .

Across all maturities and horizons, the random walk and AR(1) benchmarks achieve lower RMSE than the DNS–Kalman model. For example, at the 1-day horizon the random walk and AR(1) deliver RMSE around 0.06%, whereas DNS–Kalman is roughly twice as large; at the 21-day horizon for the 10-year yield, the random walk and AR(1) attain RMSE of approximately 0.27% and

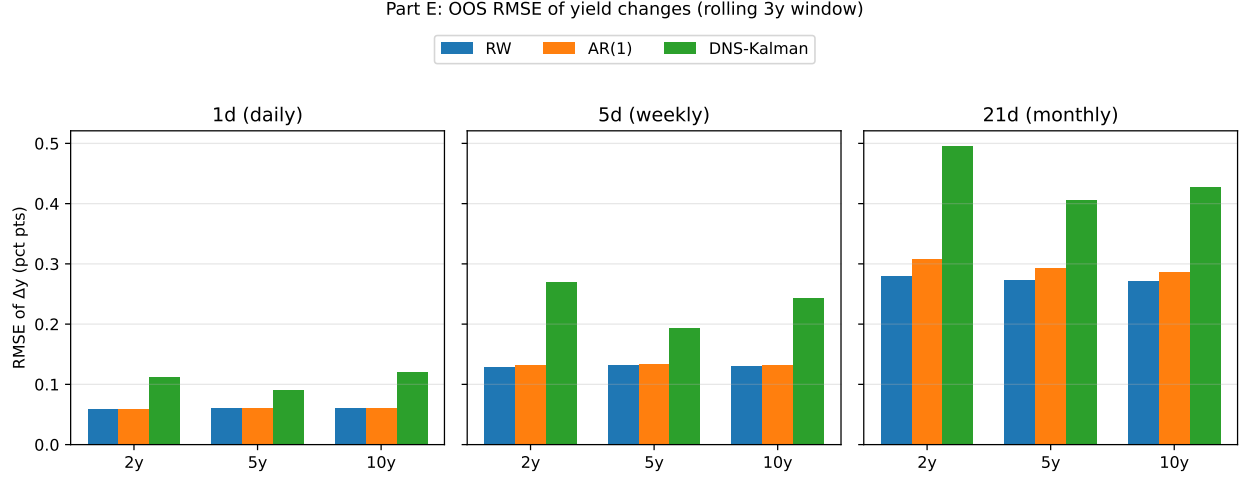


Figure 13: Out-of-sample RMSE of yield changes (%) by horizon, maturity, and model.

0.29%, compared with 0.43% for DNS–Kalman. Since expected yield changes are the primary input for duration-based trading strategies, and the DNS–Kalman forecasts do not outperform simple statistical benchmarks, we find no empirical support for DNS-driven directional trading rules and therefore do not report a P&L backtest.

6. Conclusion

We calibrate the Nelson–Siegel model to 2,524 daily U.S. Treasury yields (2016–2026) and extend it to a Kalman filter framework. Five findings emerge.

1. **Grid search and nonlinear optimisation are equivalent** (median RMSE 0.0016%); joint optimisation adds no value over a fine grid.
2. **Fixed λ fails under stress.** All methods are comparable in calm regimes, but during the 2022–2023 inversion the fixed- λ RMSE spikes up to 30 $\times$ above the adaptive methods.
3. **The Kalman filter improves factor quality.** VAR(1) dynamics eliminate OLS outliers and reduce median in-sample RMSE 10 $\times$ (0.0137% vs. 0.1296%).
4. **Filtered factors do not improve one-day *level* forecasts.** DNS–Kalman is outperformed by the random walk for 1d-ahead yields (Part D); factors remain most useful for interpolation and risk.
5. **Part E (yield *changes*):** over 1d, 5d, and 21d horizons, RW and AR(1) beat DNS–Kalman for 2y/5y/10y changes; there is no basis for DNS-driven directional trades versus simple benchmarks.

A. Closed-Form Yield Derivation

Start from

$$Y(T) = \frac{1}{T} \int_0^T \left(\beta_0 + \beta_1 e^{-\lambda s} + \beta_2 \lambda s e^{-\lambda s} \right) ds. \quad (5)$$

Compute the two nontrivial integrals:

$$\int_0^T e^{-\lambda s} ds = \left[-\frac{1}{\lambda} e^{-\lambda s} \right]_0^T = \frac{1 - e^{-\lambda T}}{\lambda}, \quad (6)$$

$$\int_0^T \lambda s e^{-\lambda s} ds = \left[-s e^{-\lambda s} \right]_0^T + \int_0^T e^{-\lambda s} ds = -T e^{-\lambda T} + \frac{1 - e^{-\lambda T}}{\lambda}. \quad (7)$$

Substituting back:

$$\begin{aligned} Y(T) &= \beta_0 + \frac{\beta_1}{T} \frac{1 - e^{-\lambda T}}{\lambda} + \frac{\beta_2}{T} \left(-T e^{-\lambda T} + \frac{1 - e^{-\lambda T}}{\lambda} \right) \\ &= \beta_0 + \beta_1 \frac{1 - e^{-\lambda T}}{\lambda T} + \beta_2 \left(\frac{1 - e^{-\lambda T}}{\lambda T} - e^{-\lambda T} \right), \end{aligned} \quad (8)$$

which gives the loadings $A_1(T; \lambda)$ and $A_2(T; \lambda)$ stated in Section 1.

For a given set of parameters, an example curve is plotted in Figure 14.

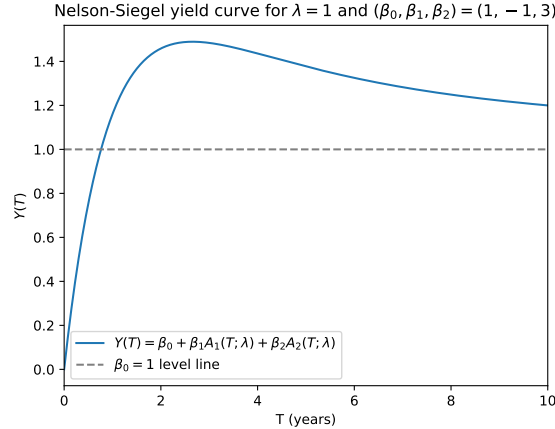


Figure 14: Example NS yield curve for $\lambda = 1$ and $(\beta_0, \beta_1, \beta_2) = (1, -1, 3)$.

B. Parameter Instability: $\beta_0 > 20$ Outliers

The fitted parameters often exhibit noisy values as a result of fitting instability: two sets of parameters can describe the same curve nearly equally well, and the model cannot distinguish between them. Since β_0 has a clear interpretation as the long-run yield level, values exceeding 20% are economically implausible and indicate fitting artefacts. We find two such dates (2023-06-29 and 2023-07-26), both during the inverted-curve regime.

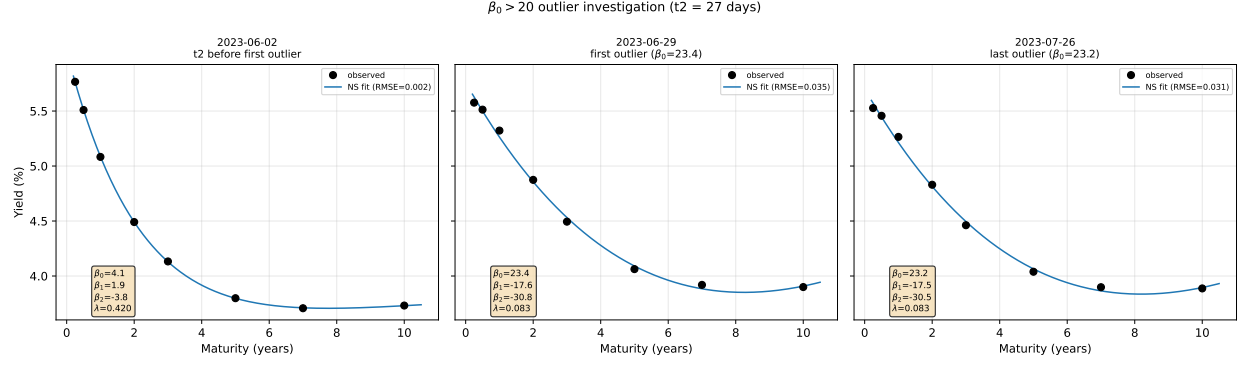


Figure 15: NS fit on two outlier dates with $\beta_0 > 20$.

For these dates, the outliers are shown in Figure 15.

On these dates the RMSE-vs- λ surface has two local minima: one near $\lambda \approx 0.08$ ($\beta_0 \approx 23$) and one near $\lambda \approx 1.6$ ($\beta_0 \approx 3.5$). The grid search selects the global minimum, producing a discontinuous jump from the $\beta_0 \approx 4$ values on neighboring days. Both solutions fit the cross-section similarly (RMSE 0.035% vs 0.039%); only the time-series context distinguishes them, which motivates the Kalman filter approach in Part D.

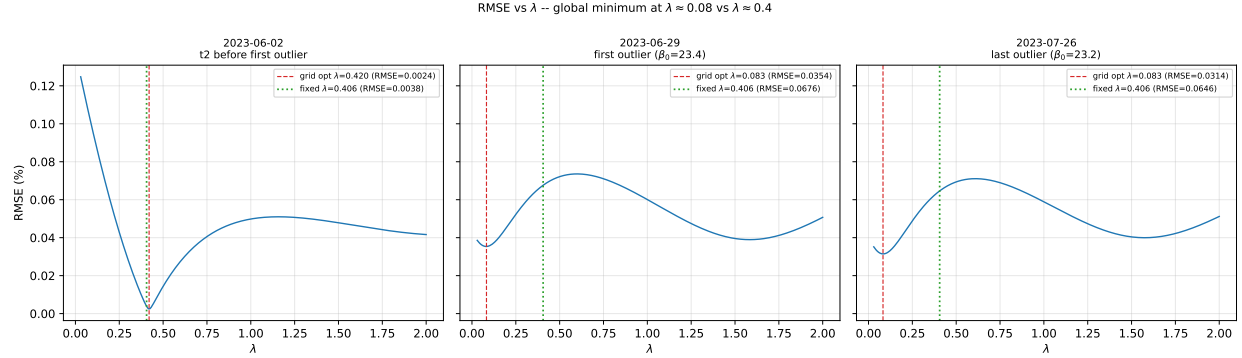


Figure 16: RMSE vs λ on the two outlier dates. The global minimum is at $\lambda \approx 0.08$ with $\beta_0 \approx 23$, while a local minimum at $\lambda \approx 1.6$ has $\beta_0 \approx 3.5$.

We test two remedies within the OLS framework to resolve this issue:

1. *Restrict to $T \geq 1$* : dropping the 0.25 Y and 0.5 Y tenors eliminates the short-end leverage that drives λ toward zero. The outliers disappear, but sub-year maturities are no longer fitted.
2. *Constrained nonlinear optimisation*: imposing $\lambda \in [0.2, 2.0]$, $\beta_0 \in [0, 10]$, $\beta_1, \beta_2 \in [-5, 5]$ keeps the parameters in an economically meaningful range at the cost of slightly higher RMSE on the outlier dates ($\approx 0.04\%$ vs $\approx 0.03\%$).

While the method "Restrict to $T \geq 1$ " solves the $\beta_0 > 20$ issue partially, and the "Constrained nonlinear optimisation" more broadly, they introduce other trade-offs, and neither addresses the underlying identification problem. The Kalman filter in Part D provides a more principled solution

by regularising the factor paths through the VAR(1) time-series structure. The outcome of these remedies is shown in Figure 17.

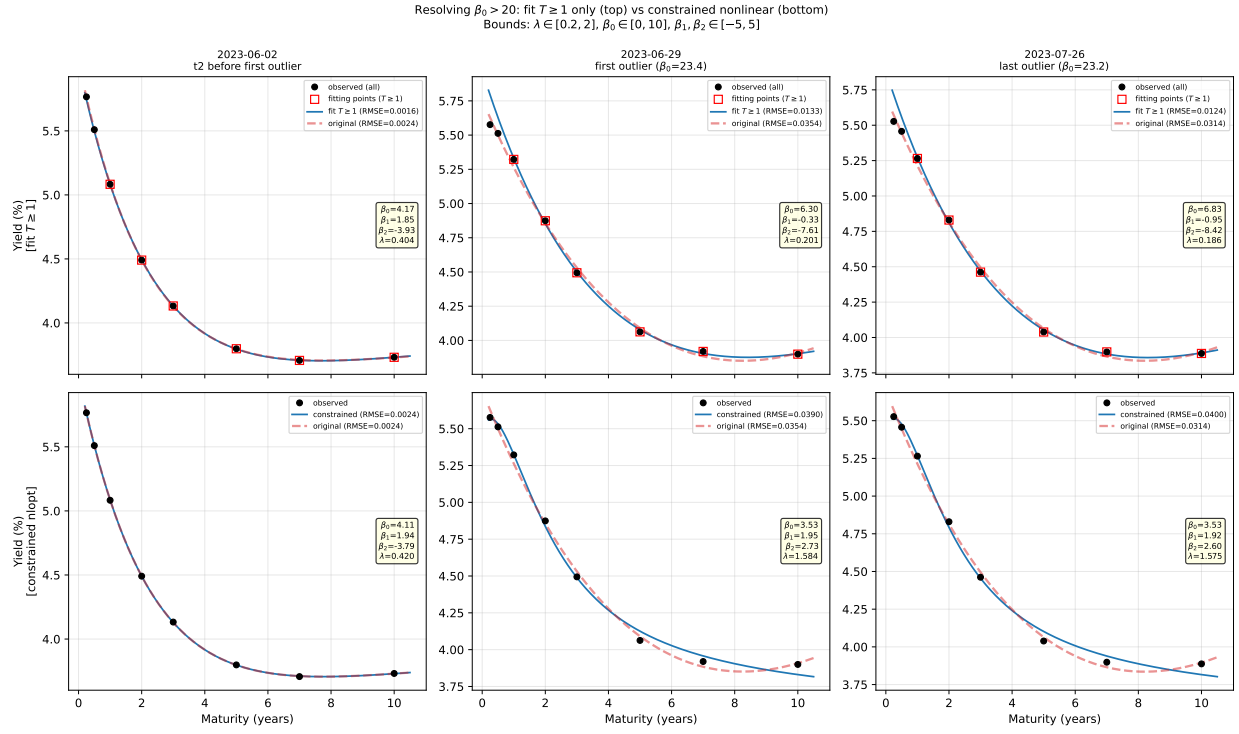


Figure 17: NS fit on the two outlier dates after applying the remedies.

To have an intuition why these remedies are not ideal, we plotted λ and β_0 as function of time in Figure 18.

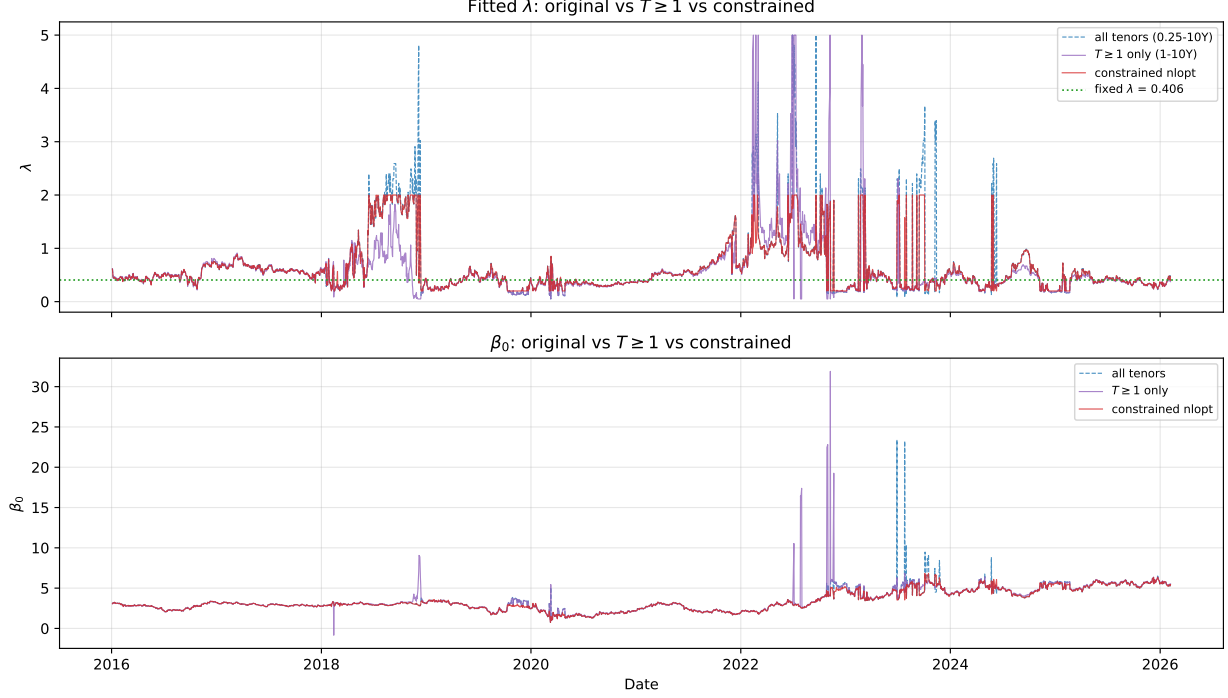


Figure 18: Time series of λ and β_0 from the grid-search method before and after applying the remedies.

C. State-Space Model Equations

The DNS state-space system consists of two equations. The **measurement equation** links observed yields to latent factors,

$$\mathbf{Y}_t = \mathbf{\Lambda} \mathbf{X}_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(\mathbf{0}, \sigma_\varepsilon^2 \mathbf{I}_8), \quad (9)$$

where $\mathbf{Y}_t \in \mathbb{R}^8$ are observed yields, $\mathbf{X}_t = (\beta_{0,t}, \beta_{1,t}, \beta_{2,t})^\top$ are the latent NS factors, and $\mathbf{\Lambda}$ is the 8×3 NS loading matrix with rows $\mathbf{\Lambda}_i = (1, \frac{1-e^{-\lambda\tau_i}}{\lambda\tau_i}, \frac{1-e^{-\lambda\tau_i}}{\lambda\tau_i} - e^{-\lambda\tau_i})$. The decay parameter is fixed at $\lambda^* = 0.406$ from Part C, making $\mathbf{\Lambda}$ a time-invariant constant matrix. The **state equation** governs factor dynamics,

$$\mathbf{X}_t = \mathbf{c} + \mathbf{\Phi} \mathbf{X}_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad (10)$$

where $\mathbf{\Phi} \in \mathbb{R}^{3 \times 3}$ is the transition matrix and $\mathbf{Q}$ is the 3×3 state-noise covariance, both estimated by multivariate OLS on the Part C daily betas.

D. Kalman Filter: Recursions and Initialisation

The Kalman filter estimates the latent factors $\mathbf{X}_t$ sequentially. At each day t we carry forward two quantities: a point estimate $\hat{\mathbf{X}}_{t|t}$ (a 3×1 vector) and its uncertainty $\mathbf{P}_{t|t}$ (a 3×3 covariance matrix). $\mathbf{P}$ encodes how confident we are in the current factor estimate: large diagonal entries mean high uncertainty. Each step consists of a *predict* phase (propagate yesterday's estimate through the VAR, which increases uncertainty by adding $\mathbf{Q}$) and an *update* phase (incorporate today's 8 observed yields, which decreases uncertainty).

Initialisation. We need starting values $\hat{\mathbf{X}}_0$ and $\mathbf{P}_0$ for the recursion. Rather than choosing these arbitrarily, we use the unconditional steady state of the VAR(1):

$$\hat{\mathbf{X}}_0 = (I_3 - \Phi)^{-1} \mathbf{c} \quad (11)$$

is the long-run mean of the factors (i.e. where the VAR would settle if run indefinitely), and $\mathbf{P}_0$ is the long-run uncertainty, obtained by solving the discrete Lyapunov equation $\mathbf{P}_0 = \Phi \mathbf{P}_0 \Phi^\top + \mathbf{Q}$. This is a matrix equation; to convert it into a standard linear system we use the vec operator (which stacks the columns of a matrix into a single vector) and the identity $\text{vec}(ABC) = (C^\top \otimes A) \text{vec}(B)$:

$$\text{vec}(\mathbf{P}_0) = (I_9 - \Phi \otimes \Phi)^{-1} \text{vec}(\mathbf{Q}). \quad (12)$$

Predict step. Before observing day t 's yields, we forecast the factors and their uncertainty from yesterday's filtered estimate:

$$\hat{\mathbf{X}}_{t|t-1} = \mathbf{c} + \Phi \hat{\mathbf{X}}_{t-1|t-1}, \quad (13)$$

$$\mathbf{P}_{t|t-1} = \Phi \mathbf{P}_{t-1|t-1} \Phi^\top + \mathbf{Q}. \quad (14)$$

The addition of $\mathbf{Q}$ reflects that the factors may have been shocked overnight; uncertainty grows.

Update step. Now we observe today's yields $\mathbf{Y}_t$ and revise:

$$\mathbf{v}_t = \mathbf{Y}_t - \Lambda \hat{\mathbf{X}}_{t|t-1} \quad (\text{innovation: how much today's yields surprised us}), \quad (15)$$

$$\mathbf{S}_t = \Lambda \mathbf{P}_{t|t-1} \Lambda^\top + \mathbf{R} \quad (\text{covariance of the innovation}), \quad (16)$$

$$\mathbf{K}_t = \mathbf{P}_{t|t-1} \Lambda^\top \mathbf{S}_t^{-1} \quad (\text{Kalman gain, } 3 \times 8), \quad (17)$$

$$\hat{\mathbf{X}}_{t|t} = \hat{\mathbf{X}}_{t|t-1} + \mathbf{K}_t \mathbf{v}_t \quad (\text{revised factor estimate}), \quad (18)$$

$$\mathbf{P}_{t|t} = (I_3 - \mathbf{K}_t \Lambda) \mathbf{P}_{t|t-1} \quad (\text{revised uncertainty — shrinks}). \quad (19)$$

Here $\mathbf{R} = \sigma_\varepsilon^2 \mathbf{I}_8$ is the measurement noise covariance from the state-space model (9). The Kalman gain $\mathbf{K}_t$ controls how much the innovation corrects the prediction. When $\mathbf{P}_{t|t-1}$ is large (uncertain prediction), $\mathbf{K}_t$ is large and the filter trusts the data. When $\mathbf{R}$ is large (noisy yields), $\mathbf{K}_t$ is small and the filter trusts the VAR prediction.

Log-likelihood. As a byproduct, the filter accumulates the log-likelihood

$$\ell = -\frac{1}{2} \sum_{t=1}^T [n \log(2\pi) + \log |\mathbf{S}_t| + \mathbf{v}_t^\top \mathbf{S}_t^{-1} \mathbf{v}_t], \quad (20)$$

which measures how “surprised” the model was by the data. We do not optimise it (we use two-step OLS estimates instead), but it could be used for full maximum-likelihood estimation.

Parameter estimation. $(\mathbf{c}, \Phi, \mathbf{Q})$ are estimated by multivariate OLS on the Part C daily betas: regress $\mathbf{X}_t$ on $[\mathbf{1}, \mathbf{X}_{t-1}]$ to obtain the 4×3 coefficient matrix, whose first row is $\mathbf{c}^\top$ and whose remaining rows form $\Phi^\top$. The residual covariance gives $\mathbf{Q}$. The measurement noise variance σ_ε^2 is set to the square of the median daily OLS RMSE from Part C, giving $\mathbf{R} = \sigma_\varepsilon^2 \mathbf{I}_8$.

E. VAR(1) Diagnostics

All eigenvalues of $\hat{\Phi}$ lie strictly inside the unit circle (Table 5), confirming stationarity. The dominant eigenvalue of 0.9958 implies a half-life of 165.7 days, reflecting the well-known near-unit-root persistence of interest rate levels. The remaining modes decay within one to five days. The unconditional means $(\bar{\beta}_0, \bar{\beta}_1, \bar{\beta}_2) = (3.62\%, -0.98\%, -1.98\%)$ are economically reasonable for the 2016–2026 sample. Impulse response functions (Appendix F) reveal strong cross-factor spillovers – particularly from slope to level – justifying the full VAR specification over independent AR(1) models.

Table 5: VAR(1) eigenvalue diagnostics

Eigenvalue	Modulus	Half-life (days)
0.9958	0.9958	165.7
0.8683	0.8683	4.9
0.5085	0.5085	1.0

F. VAR(1) Impulse Response Functions

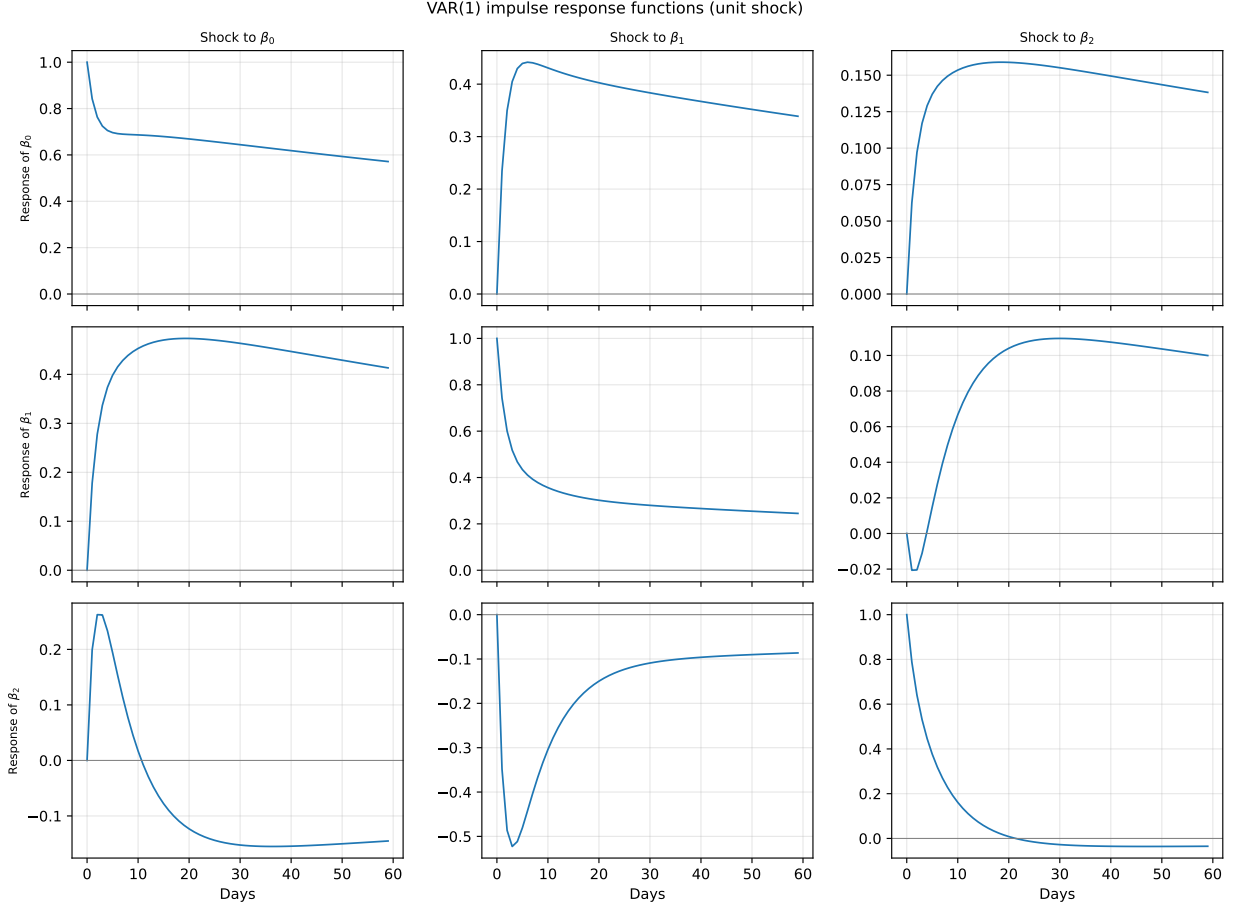


Figure 19: VAR(1) impulse response functions. Columns index the shocked factor; rows index the responding factor. No response decays to zero within 60 days owing to the dominant near-unit-root eigenvalue (half-life 166 days).

Figure 19 reports impulse response functions for a unit shock to each NS factor, traced over a 60-day horizon via iterated application of $\hat{\Phi}$.

The diagonal panels confirm that level shocks are nearly permanent (still at ≈ 0.6 after 60 days), slope shocks are moderately persistent, and curvature shocks decay the fastest. The off-diagonal panels show that slope shocks generate a persistent positive spillover into the level (reaching ≈ 0.4 and plateauing), and level shocks similarly raise the slope. These cross-factor dynamics are non-negligible and justify using the full VAR(1) rather than decoupled AR(1) models.

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